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Standard and Compound Units

Change units first. Then one triangle does speed, density and pressure: cover what you want, the other two tell you multiply or divide.

THE TOPIC ON ONE MAP

Standard and Compound Units

01 — Converting units

4 m² into cm²

32.4 m³ into cm³

50 m/s into km/h

02 — Speed, distance, time

3 h 24 min

average speed of a whole journey

03 — Density, mass, volume

mass of a gold bar

which metal?

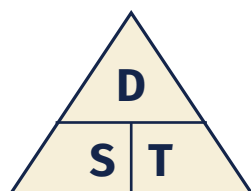
density of a mixture

04 — Pressure, force, area

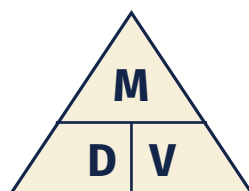
force on a table

volume of the cube

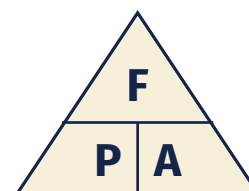
ONE TRIANGLE, THREE FORMULAS Cover the one you want:



SPEED



DENSITY



PRESSURE

ZONE AREA

m² → cm²

×100² = ×10 000

ZONE VOLUME

m³ → cm³

×100³ = ×1 000 000

ZONE SPEED

m/s → km/h

×3.6 (km/h → m/s: ÷3.6)

01 Converting Units

Length units convert once, area units twice (square the factor), volume units three times (cube it). For compound units, convert the top and the bottom separately.

WHEN YOU SEE → YOU DO

"change 4 m^2 into cm^2 "	→ $\times 100^2$
"change 32.4 m^3 into cm^3 "	→ $\times 100^3$
"50 m/s into km/h"	→ $\times 3600 \div 1000$, i.e. $\times 3.6$
"81 km/h into m/s"	→ $\div 3.6$
" w m/s in km/h, in terms of w "	→ $3.6w$

WORKED MODEL • JUNE 2022 P2H

Change 81 km/h into m/s.

1 $81 \times 1000 = 81\,000 \text{ m per hour}$

km → m

2 $81\,000 \div 3600$

hour → seconds

ANSWER 22.5 m/s

× **TRAP** $1 \text{ m}^2 = 100 \text{ cm}^2$ is wrong. A square metre is $100 \times 100 = 10\,000 \text{ cm}^2$.

IDEA 01

Practise: Converting Units

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Square the factor

CH 4 • TOPIC 1 2 marks

Change 3.5 m^2 into cm^2 .

SELECTED PRACTICE**Q2** PRACTICE Practice • Easy

CH 4 • TOPIC 1 2 marks

1 A rectangle has an area of 4 m^2 . Write this area in cm^2 .

(2 marks)

Q3**PRACTICE**

Practice • Easy

CH 4 • TOPIC 1 1 mark

14 Circle the volume that is the same as 15 cm^3

$15\,000 \text{ mm}^3$	1.5 mm^3	0.0015 mm^3	150 mm^3
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(1 mark)

Q4**PRACTICE**

Practice • Medium

CH 4 • TOPIC 1 3 marks

10 There are 18500 gallons of fuel in a fuel tank.

The fuel is pumped from the fuel tank into a plane at a rate of 1700 litres per minute.

1 gallon = 4.5 litres.

How many minutes will it take to empty the fuel tank completely? Give your answer to the nearest minute.

(3 marks)

Q5**PRACTICE**

Practice • Medium

CH 4 • TOPIC 1 1 mark

16 Which **one** of these is a unit of density? Circle your answer.

kg/m^2	m^2/kg	kg/m^3	m^3/kg
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(1 mark)

PAST PAPERS**Q6****PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 2H • Q2

CH 4 • TOPIC 1 2 marks

2 Change 32.4 m^3 into cm^3

..... cm^3

(Total for Question 2 is 2 marks)

Q7**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 2H • Q10

CH 4 • TOPIC 1**3 marks**

10 Change a speed of 50 metres per second to a speed in kilometres per hour.

..... kilometres per hour

(Total for Question 10 is 3 marks)

Q8**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2H • Q7

CH 4 • TOPIC 1**3 marks**

7 Change a speed of 81 kilometres per hour to a speed in metres per second.

..... metres per second

(Total for Question 7 is 3 marks)

Q9**PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2H • Q9

CH 4 • TOPIC 1 3 marks

- 9 Change a speed of 27 kilometres per hour to a speed in metres per second.

..... m/s

(Total for Question 9 is 3 marks)**Q10****PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2H • Q6

CH 4 • TOPIC 1 3 marks

- 6 Change a speed of w metres per second to a speed in kilometres per hour.
Give your answer in terms of w in its simplest form.

..... kilometres per hour

(Total for Question 6 is 3 marks)

Q11**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1HR • Q7

CH 4 • TOPIC 1 3 marks

7 Change a speed of $50x$ metres per second to a speed in kilometres per hour.

..... km/h

(Total for Question 7 is 3 marks)

02 Speed, Distance, Time

Minutes must become hours first ($\div 60$). Average speed = **total** distance $\div$ **total** time — never the mean of the speeds.

WHEN YOU SEE → YOU DO

“3 hours 24 minutes”

$$\rightarrow 3 + \frac{24}{60} = 3.4 \text{ hours}$$

“average speed for the whole journey”

$$\rightarrow \text{total distance} \div \text{total time}$$

“how long ... in seconds”

$$\rightarrow \text{time} = \frac{\text{distance}}{\text{speed}}, \text{ then } \times 3600$$

“show that he could run it in less than 14 minutes”

$$\rightarrow \text{scale up, compare}$$

WORKED MODEL • JANUARY 2022 P1HR

245 km in 2.5 h; 220 km at 80 km/h; then 50 minutes at 72 km/h. Average speed?

1 times: 2.5, $220 \div 80 = 2.75$, $\frac{50}{60}$ h

each part

2 distances: $245 + 220 + 60 = 525$ km

$$72 \times \frac{5}{6} = 60$$

ANSWER $525 \div 6.083 = 86.3$ km/h

× **TRAP** 3 h 24 min is not 3.24 hours. Divide the minutes by 60.

IDEA 02

Practise: Speed, Distance, Time

My questions first, then selected practice, then real past papers.

Q12 DR ESAM Minutes to hours

CH 4 • TOPIC 1 3 marks

A car travels 186 km in 2 hours 15 minutes. Work out its average speed.

SELECTED PRACTICE**Q13** PRACTICE Practice • Easy

CH 4 • TOPIC 1 4 marks

- 2 (a) Peter goes for a walk.
He walks 15 miles in 6 hours.

Work out Peter's average speed.
Give your answer in miles per hour.

(2 marks)

- (b) 5 miles = 8 km.
Sunita says that Peter walked more than 20 km.

Is Sunita right?
You must show all your working.

(2 marks)

Q14**PRACTICE**

Practice • Easy

CH 4 • TOPIC 1**5 marks**

18 (a) Hector can run 400 metres in 66 seconds.

Use this information to show that he could run 5 kilometres in less than 14 minutes.

(4 marks)

(b) Hector tries to run 5 kilometres in less than 14 minutes.

Give one reason why he might not achieve this.

(1 mark)

Q15**PRACTICE**

Practice • Medium

CH 4 • TOPIC 1 4 marks

17 Arron ran a distance of 5 km at an average speed of 2.2 m/s.

How long did Arron run for?

Give your answer in minutes and seconds, to the nearest second.

..... minutes seconds

(4 marks)

Q16**PRACTICE**

Practice • Hard

CH 4 • TOPIC 1 3 marks

- 11 (a)** In May 2019, the distance between Earth and Mars was 3.9×10^7 km.

In May 2019, a signal was sent from Earth to Mars.

Assuming that the signal sent from Earth to Mars travelled at a speed of 3×10^5 km per second, how long did the signal take to get to Mars?

.....seconds

(2 marks)

- (b)** The speed of the signal sent from Earth to Mars in May 2019 was actually less than 3×10^5 km per second.

How will this affect your answer to part (a)?

(1 mark)

PAST PAPERS

Q17**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 2HR • Q5

CH 4 • TOPIC 1 3 marks

- 5 A plane takes 3 hours 36 minutes to fly from the Cayman Islands to New York.
The plane flies a distance of 2470 km.

Work out the average speed of the plane in km/h.
Give your answer correct to the nearest whole number.

..... km/h

(Total for Question 5 is 3 marks)**Q18****PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q7

CH 4 • TOPIC 1 4 marks

- 7 Pedro drove from Toulouse to Montpellier in 2 hours 42 minutes.
He drove at an average speed of 90 km/hour.

Janine drove from Toulouse to Montpellier along the same route as Pedro.
The journey took her 3 hours.

Work out Janine's average speed for the journey.

..... km/hour

(Total for Question 7 is 4 marks)

Q19**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q6

CH 4 • TOPIC 1 3 marks

- 6 A rocket travelled 100 km at an average speed of 28 440 km/h.

Work out how long it took the rocket to travel the 100 km.

Give your answer in seconds, correct to the nearest second.

..... seconds

(Total for Question 6 is 3 marks)

Q20**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1HR • Q4

CH 4 • TOPIC 1 3 marks

- 4 A train journey from Paris to Amsterdam took 3 hours 24 minutes.
The total distance the train travelled was 433.5 km.

Work out the average speed of the train.

Give your answer in kilometres per hour.

..... km/h

(Total for Question 4 is 3 marks)

Q21**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2H • Q1

CH 4 • TOPIC 1**3 marks**

You must write down all the stages in your working.

- 1** A train takes 6 hours 39 minutes to travel from New Delhi to Kanpur.
The train travels a distance of 429 km.

Work out the average speed of the train.

Give your answer in km/h correct to one decimal place.

..... km/h

(Total for Question 1 is 3 marks)

Q22**PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 1H • Q1

CH 4 • TOPIC 1**3 marks**

You must write down all the stages in your working.

- 1** A plane flew from Madrid to Dubai.

The distance the plane flew was 5658 km.

The flight time was 8 hours 12 minutes.

Work out the average speed of the plane.

..... km/h

(Total for Question 1 is 3 marks)

Q23**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 1HR • Q7

CH 4 • TOPIC 1 4 marks

- 7 Milly went on a car journey.
She travelled from Anesey to Breigh to Clando and then to Duckbridge.

For Anesey to Breigh, Milly drove the 245 km in 2.5 hours.

For Breigh to Clando, Milly drove the 220 km at an average speed of 80 km/h

For Clando to Duckbridge, Milly drove at an average speed of 72 km/h in 50 minutes.

Work out Milly's average speed, in km/h, for the journey from Anesey to Duckbridge.

Give your answer correct to one decimal place.

..... km/h

(Total for Question 7 is 4 marks)

Q24**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2H • Q3

CH 4 • TOPIC 1 3 marks

- 3 An aeroplane travelled from New York City to Los Angeles.

The aeroplane travelled a distance of 3980 kilometres in 5 hours 24 minutes.

Work out the average speed of the aeroplane.

Give your answer in kilometres per hour correct to the nearest whole number.

..... kilometres per hour

(Total for Question 3 is 3 marks)

Q25**PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2HR • Q3

CH 4 • TOPIC 1 3 marks

- 3 Anjali travels on the Eurostar train from Paris to Amsterdam.

The distance the train travels between Paris and Amsterdam is 515 km.
The time taken by the train to travel between Paris and Amsterdam is
3 hours 18 minutes.

Work out the average speed of the train.

Give your answer in km/h correct to the nearest whole number.

..... km/h

(Total for Question 3 is 3 marks)

Q26**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2H • Q3

CH 4 • TOPIC 1 3 marks

- 3 A plane takes 9 hours 36 minutes to fly from New Delhi to Perth.

The plane flies at an average speed of 820 km/h.

Work out the total distance the plane flies.

..... km

(Total for Question 3 is 3 marks)

03 Density, Mass, Volume

Density = $\frac{\text{mass}}{\text{volume}}$. Make the units match (g with cm^3 , kg with m^3). For a mixture: total mass $\div$ total volume.

WHEN YOU SEE $\rightarrow$ YOU DO

“work out the mass of the gold bar”

$\rightarrow M = D \times V$

mass in kg, density in g/cm^3

$\rightarrow$ change kg to g first

“which metal is the solid made from?”

$\rightarrow$ find its density, match the table

“density of the alloy / mixture”

$\rightarrow$ total mass $\div$ total volume (never average the densities)

“least number of blocks”

$\rightarrow$ divide, then round up

WORKED MODEL • MIXTURE

140 g of liquid A (0.7 g/cm^3) and 128 g of liquid B (1.6 g/cm^3) are mixed. Density of the mixture?

1 $140 \div 0.7 = 200, 128 \div 1.6 = 80 \text{ cm}^3$

each volume

ANSWER $\frac{268}{280} = 0.957 \text{ g/cm}^3$

IDEA 03

Practise: Density, Mass, Volume

My questions first, then selected practice, then real past papers.

Q27 DR ESLAM Mass

CH 4 • TOPIC 1 2 marks

A block has volume 250 cm^3 and density 8.4 g/cm^3 . Work out its mass in kg.

SELECTED PRACTICE**Q28** PRACTICE Practice • Easy

CH 4 • TOPIC 1 3 marks

19 A solid metal block has mass 500 g and volume 125 cm^3 .

Work out the density of the block.

Give the units of your answer.

(3 marks)

Q29**PRACTICE**

Practice • Medium

CH 4 • TOPIC 1

3 marks

- 13** A block of wood has a mass of 3.5 kg .
The wood has density 0.65 kg/m^3

Work out the volume of the block of wood.
Give your answer correct to 3 significant figures.

(3 marks)**Q30****PRACTICE**

Practice • Hard

CH 4 • TOPIC 1

4 marks

- 4** Liquid A has a density of 0.7 g/cm^3 .
Liquid B has a density of 1.6 g/cm^3 .

140 g of liquid A and 128 g of liquid B are mixed to make liquid C.

Work out the density of liquid C.

(4 marks)

Q31**PRACTICE**

Practice · Hard

CH 4 • TOPIC 1 6 marks

13 A truck is used to transport some wood panels.

Each wood panel is a cuboid measuring 2.4 m by 1.2 m by 1.8 **cm**. The density of each wood panel is 750 kg/m^3 .

The truck can carry 15 tonnes of these wood panels.

Calculate the maximum number of wood panels that the truck can carry.
Show how you decide.

(6 marks)

PAST PAPERS

Q32 PAST PAPER Edexcel 4MA1 • January 2020 • Paper 1HR • Q9

CH 4 • TOPIC 1 5 marks

- 9 Pablo made a solid gold statue.

He melted down some gold blocks and used the gold to make the statue.
Each block of gold was a cuboid, as shown below.

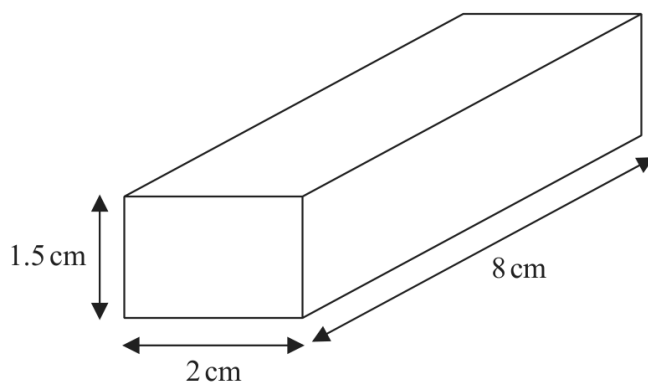


Diagram **NOT**
accurately drawn

The mass of the statue is 5.73 kg.
The density of gold is 19.32 g/cm^3

Work out the least number of gold blocks Pablo melted down in order to make the statue.
Show your working clearly.

(Total for Question 9 is 5 marks)

Q33**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 2H • Q9

CH 4 • TOPIC 1 2 marks

- 9 The density of gold is 19.3 g/cm^3
A gold bar has volume 150 cm^3

Work out the mass of the gold bar.

..... g

(Total for Question 9 is 2 marks)

Q34 **PAST PAPER** Edexcel 4MA1 · November 2021 · Paper 1H · Q10**CH 4 · TOPIC 1** **5 marks**

10 A solid aluminium cylinder has radius 10 cm and height h cm.

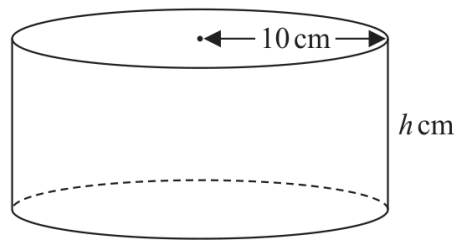


Diagram **NOT**
accurately drawn

The mass of the cylinder is 5.4 kg.

The density of aluminium is 0.0027 kg/cm^3

Calculate the value of h .

Give your answer correct to one decimal place.

$h = \dots\dots\dots$

(Total for Question 10 is 5 marks)

Q35**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2HR • Q21

CH 4 • TOPIC 1 6 marks

- 21 The diagram shows a solid made from a cylinder and a hemisphere.
The cylinder and the hemisphere are both made from the same metal.

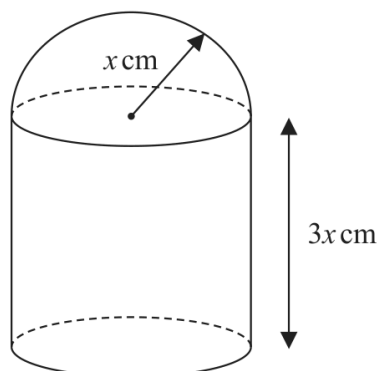


Diagram **NOT**
accurately drawn

The plane face of the hemisphere coincides with the upper plane face of the cylinder.

The radius of the cylinder and the radius of the hemisphere are both x cm.

The height of the cylinder is $3x$ cm.

The total surface area of the solid is $81\pi\text{ cm}^2$

The mass of the solid is 840 grams.

The following table gives the density of each of four metals.

Metal	Density (g/cm^3)
Aluminium	2.7
Nickel	8.9
Gold	19.3
Silver	10.5

The metal used to make the solid is one of the metals in the table.

Determine the metal used to make the solid.

Show your working clearly.

(Total for Question 21 is 6 marks)

Q36 **PAST PAPER** Edexcel 4MA1 • June 2022 • Paper 1HR • Q7**CH 4 • TOPIC 1** **3 marks**

- 7 The diagram shows a solid cylinder made from iron.

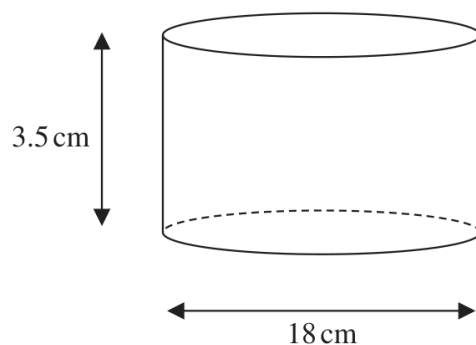


Diagram **NOT**
accurately drawn

The cylinder has diameter 18 cm and height 3.5 cm

The mass of the cylinder is 7.04 kg

Work out the density of the iron.

Give your answer in g/cm^3 correct to 2 significant figures.

..... g/cm^3

(Total for Question 7 is 3 marks)

Q37**PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1HR • Q11

CH 4 • TOPIC 1 3 marks

- 11 The diagram shows a block of iron in the shape of a cuboid.

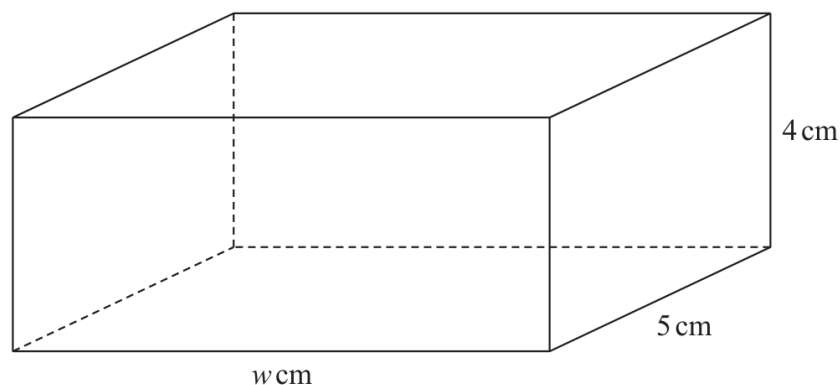


Diagram **NOT**
accurately drawn

The block has length w cm, width 5 cm and height 4 cm

The density of iron is 7.8 g/cm^3

The mass of the block is 1950 g

Work out the value of w

$w = \dots\dots\dots$

(Total for Question 11 is 3 marks)

Q38**PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1HR • Q6

CH 4 • TOPIC 1 4 marks

- 6 The diagram shows a solid wooden cylinder.

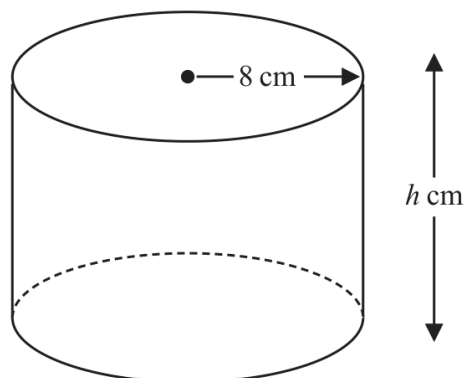


Diagram **NOT**
accurately drawn

The cylinder has radius 8 cm and height h cm.
The volume of the cylinder is 1208 cm^3

- (a) Work out the value of h
Give your answer correct to the nearest whole number.

$h = \dots\dots\dots$

The density of the wood is 1.25 g/cm^3

- (b) Work out the mass of the cylinder.
Give your answer in kilograms.

$\dots\dots\dots$ kilograms

(Total for Question 6 is 4 marks)

Q39**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2HR • Q7

CH 4 • TOPIC 1 2 marks

- 7 The mass of a silver coin is 48.3 g
The density of silver is 10.5 g/cm^3

Work out the volume of the silver coin.

..... cm^3

(Total for Question 7 is 2 marks)

Q40**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 2H • Q26

CH 4 • TOPIC 1 6 marks

26 A solid cone is joined to a solid hemisphere to make the solid shown below.

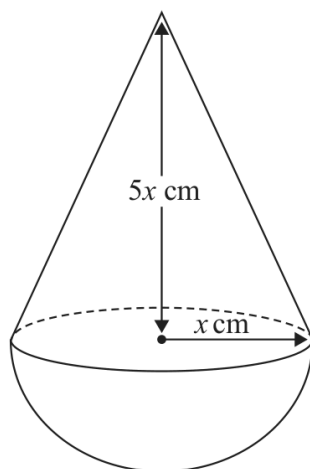


Diagram **NOT**
accurately drawn

The cone is made from copper.

The density of copper is 9 g/cm^3

The hemisphere is made from a different metal.

The total mass of the solid is 4752π grams

The total volume of the solid is $504\pi \text{ cm}^3$

Work out the density of the hemisphere.

Show your working clearly.

..... g/cm^3

(Total for Question 26 is 6 marks)

04 Pressure, Force, Area

Pressure = $\frac{\text{force}}{\text{area}}$. The area is only the face **touching** the surface. Watch cm^2 against m^2 .

WHEN YOU SEE → YOU DO

“force exerted by the box”

→ $F = P \times A$

“work out the volume of the cube”

→ $\text{area} = F \div P$, $\text{side} = \sqrt{\text{area}}$, cube it

“area 120 cm^2 ... answer in N/m^2 ”

→ $120 \text{ cm}^2 = 0.012 \text{ m}^2$

“the pressure decreases by less than 20%?”

→ work out both pressures, then the % change

WORKED MODEL • JANUARY 2020 P1H

A cube exerts 56 N; the pressure is 0.14 N/cm^2 . Work out the volume of the cube.

1 $A = 56 \div 0.14 = 400 \text{ cm}^2$

area of the face

2 $\text{side} = \sqrt{400} = 20 \text{ cm}$

square face

ANSWER $V = 20^3 = 8000 \text{ cm}^3$

IDEA 04

Practise: Pressure, Force, Area

My questions first, then selected practice, then real past papers.

Q41 DR ESLAM Force

CH 4 • TOPIC 1 2 marks

A box on a 0.8 m by 0.5 m face gives a pressure of 150 N/m^2 . Work out the force.

SELECTED PRACTICE
Q42 PRACTICE Practice • Easy

CH 4 • TOPIC 1 3 marks

5

 $F = \text{force}$
 $A = \text{area}$
 $A = \text{area}$

A box exerts a force of 140 newtons on a table.

The pressure on the table is 35 newtons/m^2 .

Calculate the area of the box that is in contact with the table.

(3 marks)

Q43**PRACTICE**

Practice • Easy

CH 4 • TOPIC 1 3 marks

11

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

A box is put on a table.

The face of the box in contact with the table is in the shape of a rectangle, 2 m by 1.25 m.

The pressure on the table due to the box is 42 newtons/m²

Work out the force exerted by the box on the table.

..... newtons

(3 marks)

Q44**PRACTICE**

Practice • Medium

CH 4 • TOPIC 1

3 marks

- 7 A force of 70 newtons acts on an area of 20 cm^2 .

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

The force is increased by 10 newtons.

The area is increased by 10 cm^2 .

Helen says,

"The pressure decreases by less than 20%"

Is Helen correct?

You must show how you get your answer.

(3 marks)

Q45**PRACTICE**

Practice • Medium

CH 4 • TOPIC 1 2 marks

8 $\text{Pressure} = \frac{\text{force}}{\text{area}}$

Find the pressure exerted by a force of 900 newtons on an area of 60cm^2 .
Give your answer in newtons/m^2 .

(2 marks)

PAST PAPERS

Q46**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 1H • Q13

CH 4 • TOPIC 1 4 marks**13** The diagram shows a solid cube.

The cube is placed on a table so that the whole of one face of the cube is in contact with the table.

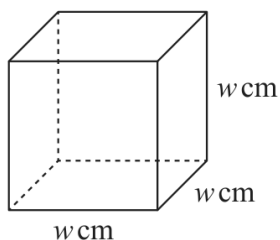


Diagram **NOT**
accurately drawn

The cube exerts a force of 56 newtons on the table.

The pressure on the table due to the cube is 0.14 newtons/cm²

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the volume of the cube.

..... cm³

(Total for Question 13 is 4 marks)

Q47**PAST PAPER**

Edexcel 4MA1 · January 2020 · Paper 2HR · Q9

CH 4 · TOPIC 1 3 marks

- 9 The diagram shows a box in the shape of a cuboid.

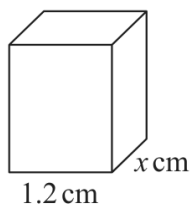


Diagram **NOT**
accurately drawn

The box is put on a table.

The face of the box in contact with the table has length 1.2 metres and width x metres.

The force exerted by the box on the table is 27 newtons.

The pressure on the table due to the box is 30 newtons/m²

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the value of x .

$x =$

(Total for Question 9 is 3 marks)

Q48**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q9

CH 4 • TOPIC 1 3 marks

- 9 A box is put on a horizontal table.

The face of the box in contact with the table is a square of side 1.5 metres.

The pressure on the table due to the box is 34.8 newtons/m²

Work out the force exerted by the box on the table.

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

..... newtons

(Total for Question 9 is 3 marks)

Q49**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1H • Q7

CH 4 • TOPIC 1 4 marks

- 7 A cylinder is placed on the ground.

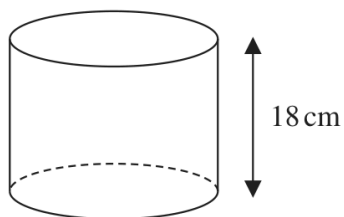


Diagram **NOT**
accurately drawn

The height of the cylinder is 18 cm.

The force exerted by the cylinder on the ground is 72 newtons.

The pressure on the ground due to the cylinder is 1.4 newtons/cm²

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the volume of the cylinder.

Give your answer correct to 3 significant figures.

..... cm³

(Total for Question 7 is 4 marks)

Q50 PAST PAPER Edexcel 4MA1 • May 2024 • Paper 1H • Q8

CH 4 • TOPIC 1 3 marks

- 8 A cylinder is placed on a table.

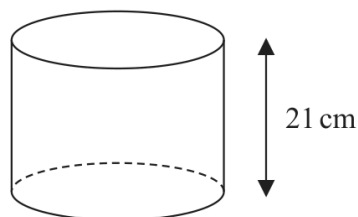


Diagram **NOT**
accurately drawn

The volume of the cylinder is 1575 cm^3

The force exerted by the cylinder on the table is 84 newtons.

$$\left| \text{pressure} = \frac{\text{force}}{\text{area}} \right|$$

Work out the pressure on the table due to the cylinder.

..... newtons/cm²

(Total for Question 8 is 3 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|-----------------|--|
| 01 | Convert | Area: square the factor; volume: cube it; m/s $\times 3.6$ = km/h. |
| 02 | Speed | Minutes $\div 60$; average = total distance $\div$ total time. |
| 03 | Density | $D = M \div V$; matching units; mixtures: total $\div$ total. |
| 04 | Pressure | $P = F \div A$; only the face in contact. |

BOTTOM LINE Units first → cover what you want in the triangle → round only at the end.

Answers

Standard and Compound Units • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • CONVERTING UNITS

Q1	Dr Eslam • Square the factor	35 000 cm ²
Q2	Practice • Easy	40 000 cm ²
Q3	Practice • Easy	15 000 mm ³
Q4	Practice • Medium	49 minutes
Q5	Practice • Medium	kg/m ³
Q6	Edexcel 4MA1 • January 2020 • Paper 2H • Q2	32 400 000 cm ³
Q7	Edexcel 4MA1 • January 2020 • Paper 2H • Q10	180 km/h
Q8	Edexcel 4MA1 • June 2022 • Paper 2H • Q7	22.5 m/s
Q9	Edexcel 4MA1 • June 2023 • Paper 2H • Q9	7.5 m/s
Q10	Edexcel 4MA1 • June 2024 • Paper 2H • Q6	$3.6w \text{ km/h} \left(\frac{18w}{5} \right)$
Q11	Edexcel 4MA1 • June 2025 • Paper 1HR • Q7	180x km/h

IDEA 02 • SPEED, DISTANCE, TIME

- Q12** Dr Eslam • Minutes to hours $186 \div 2.25 = 82.7 \text{ km/h}$
- Q13** Practice • Easy (a) 2.5 mph (b) 15 miles = 24 km > 20: yes
- Q14** Practice • Easy (a) $5000 \div 400 \times 66 = 825 \text{ s} = 13.75 \text{ min}$ (b) he may tire and slow down
- Q15** Practice • Medium 37 min 53 s
- Q16** Practice • Hard (a) 130 s (b) it would take longer
- Q17** Edexcel 4MA1 • January 2020 • Paper 2HR • Q5 686 km/h
- Q18** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q7 81 km/h
- Q19** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q6 13 seconds
- Q20** Edexcel 4MA1 • January 2021 • Paper 1HR • Q4 127.5 km/h
- Q21** Edexcel 4MA1 • January 2021 • Paper 2H • Q1 64.5 km/h
- Q22** Edexcel 4MA1 • May 2021 • Paper 1H • Q1 690 km/h
- Q23** Edexcel 4MA1 • January 2022 • Paper 1HR • Q7 86.3 km/h
- Q24** Edexcel 4MA1 • June 2022 • Paper 2H • Q3 737 km/h
- Q25** Edexcel 4MA1 • June 2023 • Paper 2HR • Q3 156 km/h
- Q26** Edexcel 4MA1 • June 2024 • Paper 2H • Q3 7872 km

IDEA 03 • DENSITY, MASS, VOLUME

Q27	Dr Eslam • Mass	2100 g = 2.1 kg
Q28	Practice • Easy	4 g/cm ³
Q29	Practice • Medium	5.38 m ³
Q30	Practice • Hard	$\frac{268}{280} = 0.957 \text{ g/cm}^3$
Q31	Practice • Hard	each panel 38.88 kg: 385 panels
Q32	Edexcel 4MA1 • January 2020 • Paper 1HR • Q9	each block 463.68 g: 13 blocks
Q33	Edexcel 4MA1 • January 2020 • Paper 2H • Q9	2895 g
Q34	Edexcel 4MA1 • November 2021 • Paper 1H • Q10	$h = 6.4$
Q35	Edexcel 4MA1 • January 2022 • Paper 2HR • Q21	$9\pi x^2 = 81\pi$, $x = 3$; $V = 99\pi$: density 2.70 — aluminium
Q36	Edexcel 4MA1 • June 2022 • Paper 1HR • Q7	7.9 g/cm ³
Q37	Edexcel 4MA1 • May 2023 • Paper 1HR • Q11	$w = 12.5$
Q38	Edexcel 4MA1 • May 2024 • Paper 1HR • Q6	(a) $h = 6$ (b) 1.51 kg
Q39	Edexcel 4MA1 • June 2025 • Paper 2HR • Q7	4.6 cm ³
Q40	Edexcel 4MA1 • November 2025 • Paper 2H • Q26	$x = 6$; hemisphere 1512π g in $144\pi \text{ cm}^3$: 10.5 g/cm ³

IDEA 04 • PRESSURE, FORCE, AREA

Q41	Dr Eslam • Force	$150 \times 0.4 = 60 \text{ N}$
Q42	Practice • Easy	4 m^2
Q43	Practice • Easy	105 N
Q44	Practice • Medium	$3.5 \rightarrow 2.67 \text{ N/cm}^2$: 23.8% decrease — Helen is wrong
Q45	Practice • Medium	$150\,000 \text{ N/m}^2$
Q46	Edexcel 4MA1 • January 2020 • Paper 1H • Q13	8000 cm^3
Q47	Edexcel 4MA1 • January 2020 • Paper 2HR • Q9	$x = 0.75$
Q48	Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q9	78.3 N
Q49	Edexcel 4MA1 • June 2022 • Paper 1H • Q7	926 cm^3
Q50	Edexcel 4MA1 • May 2024 • Paper 1H • Q8	1.12 N/cm^2